
Welcome,
LIGHT
DETECTIVES!

Our Mystery Question

How can we see things?









What do you notice?

What looks different?

Which room is easiest to see the bear?

Day 1

Light Detectives

Meet Our Light Sources!

A light source gives off its own light.



Sun

The sun gives off light.



Lamp

A lamp gives off light.



Flashlight

A flashlight gives off light.



Candle

A candle gives off light.

LIGHT SOURCES GIVE OFF LIGHT!

Which of these are light sources?

Day 1
Light Detectives

How Does Light Help Us See?

Light travels from a light source to an object and then bounces to our eyes. This helps us see the object.

Light helps us see...

Colors



Shapes



Details



What helps you see the color, shape, and details of this bear?

Day 1
Light Detectives

Bright, Dim, or Dark?

The amount of light can change how well we see objects.

Lots of Light



We can see the bear clearly.
We can see its color, shape, and details.

Little Light



We can still see the bear,
but it is harder to see the details.

Very Little or No Light



It is very hard or impossible
to see the bear.

What do you notice?

The Mystery Box Experiment!

Day 1
Light Detectives

Let's investigate how light helps us see objects!

What We Need:


Small cardboard box


Small toy (or object)


Flashlight

What do you think is inside the box?

Will we be able to see the toy without light? With a flashlight?

Make a Prediction!


I think we will see the toy without light.

Yes

No


I think we will see the toy with a flashlight.

Yes

No

Day 1
Light Detectives

Test the Mystery Box!

Follow the steps to see how light helps us see the toy.



1 Look Inside with Little Light



Look inside the box with the lid closed or a small opening. What do you see?

2 Shine the Flashlight



Shine the flashlight into the box. What happens?

3 Observe the Toy



Look closely at the toy. What can you see now?

4 Compare and Discuss



How is it different? Why do you think the flashlight helped?

With Little or No Light



It is hard to see the toy. The colors and details are not clear.

Compare!

With a Flashlight



The toy is easier to see. We can see its color, shape, and details.

Day 1
Light Detectives

What Did We Discover?

More light helps us see objects more clearly!

Without the Flashlight



With the Flashlight





What We Learned

- Light helps us see objects.
- With more light, we can see the colors, shapes, and details.
- Without light, it is hard or impossible to see objects.

Day 1
Light Detectives

Light Source or Not?

Does it give off its own light?



Sun

Light Source | Not a Light Source



Lamp

Light Source | Not a Light Source



Candle

Light Source | Not a Light Source



Flashlight

Light Source | Not a Light Source



Moon

Light Source | Not a Light Source



TV Screen

Light Source | Not a Light Source



Mirror

Light Source | Not a Light Source



Book

Light Source | Not a Light Source



Phone

Light Source | Not a Light Source



Rock

Light Source | Not a Light Source





Which objects give off their own light?

Which objects do not give off their own light?

Day 1
Light Detectives

Let's Sort!

Look at each picture. Decide if it is a light source or not a light source. Drag it to the correct box!

Does it give off its own light?

Light Source

Not a Light Source







Day 1
Light Detectives

What We Learned Today!

We learned so much about light!

Light helps us see objects.



We can see colors, shapes, and details when there is light.

A light source gives off its own light.



The sun, a lamp, a flashlight, and a candle are light sources.

Some things are not light sources.



Toys, books, rocks, trees, and people do not give off their own light.

More light helps us see more clearly.



The more light there is, the easier it is to see the colors, shapes, and details.

We can identify light sources and non-light sources.



We can look at an object and decide if it gives off its own light or if it is not a light source.

